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Unphysicality of Hilbert spaces for quantum mechanics

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Category: Quantum mechanics - Tags: Hilbert spaces, Functional analysis

Abstract

We show that Hilbert spaces cannot faithfully represent the set of states for a physical system. In particular, since observables are not guaranteed to be continuous and well-defined over the whole space, small changes in state can correspond to large changes in expectation values and states cannot be equivalently expressed in all frames. Moreover, since all separable infinite-dimensional Hilbert spaces are isomorphic, the state spaces for systems with different numbers of particles are indistinguishable.

1 Introduction

In this note we present some problems with using Hilbert spaces to represent states in quantum mechanics: small changes in state do not lead to small changes of observables (i.e. unbounded operators cannot be continuous), observables are not well defined in all frames and the state spaces for systems with different numbers of particles are indistinguishable.[1]

While these problems are largely known in the mathematical community, they are a surprise to most physicists, as they usually do not impact calculations in physically meaningful cases. They, therefore, represent obstacles to a crisp understanding of the theory, not to the application of the theory to experimental settings.

2 Mathematical prerequisites

The theorem relies on standard mathematical results and examples that showcase those results. The proofs are not particularly physically interesting, and their physical significance is discussed in the proof of the main theorem.

Proposition 1. *Let $\mathcal{H}$ be a complex Hilbert space and let $A : D(A) \subseteq \mathcal{H} \to \mathcal{H}$ be an unbounded self-adjoint operator. Then $D(A) \neq \mathcal{H}$, and A is not continuous when $D(A)$ carries the norm inherited from $\mathcal{H}$.*

Proof. This proposition collects two standard results from functional analysis; we include a proof for completeness.

[1] If the reader is aware of other problems that should be added, do contact us.

First, suppose that $D(A) = \mathcal{H}$. We show that the graph of A is closed. Let $x_n \to x$ and $Ax_n \to y$ in $\mathcal{H}$. Since A is self-adjoint, for every $z \in \mathcal{H}$,

$$\langle z, y\rangle = \lim_{n\to\infty} \langle z, Ax_n\rangle = \lim_{n\to\infty} \langle Az, x_n\rangle = \langle Az, x\rangle = \langle z, Ax\rangle.$$

Thus $\langle z, y - Ax\rangle = 0$ for every z. Taking $z = y - Ax$ gives $\|y - Ax\|^2 = 0$, so $y = Ax$ and the graph is closed. The closed graph theorem now implies that A is bounded, contradicting the hypothesis. Therefore $D(A) \neq \mathcal{H}$.

To prove discontinuity, unboundedness allows us to choose, for every positive integer n, a vector $u_n \in D(A)$ with $\|u_n\| = 1$ and $\|Au_n\| > n$. Define

$$y_n = \frac{u_n}{\|Au_n\|}.$$

Since $D(A)$ is a vector space, $y_n \in D(A)$. Moreover,

$$\|y_n\| = \frac{\|u_n\|}{\|Au_n\|} = \frac{1}{\|Au_n\|} < \frac{1}{n} \longrightarrow 0,$$
$$\|Ay_n\| = \left\|A\left(\frac{u_n}{\|Au_n\|}\right)\right\| = \frac{\|Au_n\|}{\|Au_n\|} = 1.$$

Thus $y_n \to 0$ but $Ay_n \not\to A0 = 0$. Hence A is not continuous at zero. □

Proposition 2. *Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$ with orthonormal basis $(e_n)_{n\geq 0}$. Define*

$$D(N) = \left\{\sum_{n=0}^{\infty} a_n e_n \in \mathcal{H} : \sum_{n=0}^{\infty} n^2|a_n|^2 < \infty\right\},$$
$$N\left(\sum_{n=0}^{\infty} a_n e_n\right) = \sum_{n=0}^{\infty} n a_n e_n.$$

Then N is an unbounded self-adjoint operator, and its expectation function

$$E_N(x) = \frac{\langle x, Nx\rangle}{\|x\|^2}, \qquad x \in D(N) \setminus \{0\},$$

is not continuous in the Hilbert norm, even when restricted to normalized vectors.

Proof. The domain contains all finite linear combinations of the basis vectors, so it is dense. For $x, y \in D(N)$,

$$\langle x, Ny\rangle = \sum_{n=0}^{\infty} \overline{x_n}\, n y_n = \sum_{n=0}^{\infty} \overline{n x_n}\, y_n = \langle Nx, y\rangle.$$

Thus N is symmetric. Conversely, if $z \in D(N^*)$ and $h = N^* z$, testing the adjoint identity on e_n gives

$$n\overline{z_n} = \langle z, Ne_n\rangle = \langle h, e_n\rangle = \overline{h_n}.$$

Hence $h_n = n z_n$. Since $h \in \mathcal{H}$, $\sum_n n^2|z_n|^2 = \|h\|^2 < \infty$, so $z \in D(N)$ and $N^* z = Nz$. Therefore $N^* = N$. Moreover, $\|Ne_n\| = n$ while $\|e_n\| = 1$, so N is unbounded.

For every positive integer j, define

$$x_j = \sqrt{1 - \frac{1}{j}}\, e_0 + \sqrt{\frac{1}{j}}\, e_j.$$

These finite linear combinations belong to $D(N)$, and orthonormality gives

$$\begin{aligned} \|x_j\|^2 &= 1 - \frac{1}{j} + \frac{1}{j} = 1, \\ \|x_j - e_0\|^2 &= \left(\sqrt{1 - \frac{1}{j}} - 1\right)^2 + \frac{1}{j} \\ &= 2 - 2\sqrt{1 - \frac{1}{j}} \longrightarrow 0. \end{aligned}$$

However, $Ne_0 = 0$ and

$$Nx_j = \sqrt{1 - \frac{1}{j}}\, Ne_0 + \sqrt{\frac{1}{j}}\, Ne_j = j\sqrt{\frac{1}{j}}\, e_j.$$

Therefore,

$$E_N(x_j) = \langle x_j, Nx_j \rangle = j\sqrt{1 - \frac{1}{j}}\sqrt{\frac{1}{j}}\, \langle e_0, e_j \rangle + j\left(\sqrt{\frac{1}{j}}\right)^2 \langle e_j, e_j \rangle = 0 + j\frac{1}{j} = 1,$$

whereas $E_N(e_0) = \langle e_0, Ne_0 \rangle = 0$. Thus $x_j \to e_0$ but $E_N(x_j) \not\to E_N(e_0)$, which proves the claimed discontinuity. □

Proposition 3. *Let $\mathcal{H}_x = L^2(\mathbb{R}, dx)$ and $\mathcal{H}_y = L^2(\mathbb{R}, dy)$. Let X and Y be the multiplication operators*

$$\begin{aligned} (X\psi)(x) &= x\psi(x), \quad D(X) = \{\psi \in \mathcal{H}_x : x\psi(x) \in \mathcal{H}_x\}, \\ (Y\phi)(y) &= y\phi(y), \quad D(Y) = \{\phi \in \mathcal{H}_y : y\phi(y) \in \mathcal{H}_y\}. \end{aligned}$$

The function

$$f(x) = \tan\left(\frac{\pi}{2}\operatorname{erf}(x)\right)$$

is a smooth increasing bijection of $\mathbb{R}$. Writing $g = f^{-1}$, the map

$$(U\psi)(y) = \psi(g(y))\sqrt{g'(y)}$$

is unitary from $\mathcal{H}_x$ to $\mathcal{H}_y$ and maps the normalized function

$$\psi(x) = \sqrt{\frac{e^{-x^2}}{\sqrt{\pi}}} \quad \text{to} \quad \phi(y) = \sqrt{\frac{1}{\pi(1 + y^2)}}.$$

However, $\psi \in D(X)$ while $\phi \notin D(Y)$. The transported operator UXU^{-1} is the maximal multiplication operator by $g(y)$, with domain $UD(X)$.

Proof. Since $\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$,

$$f'(x) = \frac{\pi}{2} \frac{2}{\sqrt{\pi}} e^{-x^2} \sec^2\left(\frac{\pi}{2} \operatorname{erf}(x)\right) = \sqrt{\pi} e^{-x^2} \left(1 + f(x)^2\right) > 0.$$

Moreover, $\operatorname{erf}(x) \to \pm 1$ as $x \to \pm\infty$, so $f(x) \to \pm\infty$. Thus f is a smooth increasing bijection with smooth inverse g, and

$$g'(y) = \frac{1}{f'(g(y))} = \frac{e^{g(y)^2}}{\sqrt{\pi}(1 + y^2)}.$$

The change of variables $x = g(y)$ gives

$$\|U\psi\|^2 = \int_{\mathbb{R}} |\psi(g(y))|^2 g'(y)\, dy = \int_{\mathbb{R}} |\psi(x)|^2\, dx = \|\psi\|^2.$$

Define

$$(V\phi)(x) = \phi(f(x)) \sqrt{f'(x)}.$$

Similarly,

$$\|V\phi\|^2 = \int_{\mathbb{R}} |\phi(f(x))|^2 f'(x)\, dx = \int_{\mathbb{R}} |\phi(y)|^2\, dy.$$

Using $g(f(x)) = x$, $f(g(y)) = y$ and the derivatives of these identities, we obtain

$$(VU\psi)(x) = \psi(g(f(x))) \sqrt{g'(f(x)) f'(x)} = \psi(x),$$
$$(UV\phi)(y) = \phi(f(g(y))) \sqrt{f'(g(y)) g'(y)} = \phi(y).$$

Therefore U is a linear isometry with inverse V, and hence is unitary.

The Gaussian is normalized because

$$\|\psi\|^2 = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-x^2}\, dx = 1.$$

Its image is

$$(U\psi)(y) = \sqrt{\frac{e^{-g(y)^2}}{\sqrt{\pi}}} \sqrt{\frac{e^{g(y)^2}}{\sqrt{\pi}(1 + y^2)}} = \sqrt{\frac{1}{\pi(1 + y^2)}} = \phi(y).$$

Unitarity implies $\|\phi\| = 1$. Integration by parts gives

$$\|X\psi\|^2 = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} x^2 e^{-x^2}\, dx = -\frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} x \frac{d}{dx} e^{-x^2}\, dx = -\frac{1}{2\sqrt{\pi}} \left[x e^{-x^2}\right]_{-\infty}^{\infty} + \frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} e^{-x^2}\, dx = \frac{1}{2}.$$

Thus $\psi \in D(X)$. In contrast,

$$\int_{\mathbb{R}} y^2 |\phi(y)|^2\, dy = \frac{2}{\pi} \lim_{R \to \infty} \int_0^R \frac{y^2}{1 + y^2}\, dy = \frac{2}{\pi} \lim_{R \to \infty} (R - \arctan R) = \infty.$$

Hence $\phi \notin D(Y)$.

Finally, for $\eta \in \mathcal{H}_x$, change of variables gives

$$\int_{\mathbb{R}} |g(y)(U\eta)(y)|^2\, dy = \int_{\mathbb{R}} x^2 |\eta(x)|^2\, dx.$$

Thus $gU\eta \in \mathcal{H}_y$ if and only if $\eta \in D(X)$, establishing the asserted domain. On this domain,

$$(UX\eta)(y) = g(y)\eta(g(y))\sqrt{g'(y)} = g(y)(U\eta)(y).$$

Therefore UXU^{-1} is the maximal multiplication operator by $g(y)$. □

We recall a standard classification result for Hilbert spaces, with a short proof for completeness.

Proposition 4. *Any two separable infinite-dimensional complex Hilbert spaces are unitarily isomorphic. In particular, $L^2(\mathbb{R}^n)$ and $L^2(\mathbb{R}^k)$ are unitarily isomorphic for all positive integers n, k.*

Proof. Let $(e_j)_{j\geq 0}$ and $(f_j)_{j\geq 0}$ be countable orthonormal bases of $\mathcal{H}$ and $\mathcal{K}$. Define

$$U\left(\sum_{j=0}^{\infty} a_j e_j\right) = \sum_{j=0}^{\infty} a_j f_j.$$

By Parseval's identity, both series converge precisely when $\sum_j |a_j|^2 < \infty$. Thus U is well-defined, linear, and has inverse given by exchanging the two bases. For $x = \sum_j a_j e_j$ and $y = \sum_j b_j e_j$,

$$\langle Ux, Uy\rangle = \sum_{j=0}^{\infty} \overline{a_j} b_j = \langle x, y\rangle.$$

Hence U is unitary. Since $L^2(\mathbb{R}^n)$ is separable and infinite-dimensional for every positive integer n, the particular case follows. □

3 Conditions

The following are conditions that we would expect any physical theory to obey, and therefore they should be considered part of a base theory for all of physics.

Condition [B]COBS (Continuity of measurable quantities)**.** *Small state changes lead to small changes of measurable quantities. At a minimum, the expectation of a quantity is a continuous function of the state.*

Condition [B]WDQ (Measurable quantities well-defined in all frames)**.** *All states can equivalently be expressed in all frames. At a minimum, if a state has finite variance for position in one frame, it should also have finite variance in position in another frame.*

Condition [B]STDIST (Distinguishability of different systems)**.** *Systems with a different number of degrees of freedom are distinct.*

The following condition sums up the standard definition of quantum states and observables on top of Hilbert spaces. We will test this against the above requirements.

Condition QSTO:H (Quantum states and observables)**.** *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy $S(\rho) = -\operatorname{tr}(\rho\log\rho)$.*

4 Theorem

In this theorem, we want to check whether [B]COBS, [B]WDQ and [B]STDIST are guaranteed by the representation provided by the background condition. It turns out they aren't.

Theorem 5. *Over QSTO:H, conditions [B]COBS, [B]WDQ and [B]STDIST are not guaranteed.*

Proof. First, let us show that [B]COBS is not guaranteed over QSTO:H. Condition QSTO:H is applicable to all quantum systems; therefore, it has to apply to a standard one-dimensional harmonic oscillator. Let N be the standard number operator. This satisfies the conditions for 2. If $|i\rangle$ are the energy levels, $\sqrt{\frac{j-1}{j}}|0\rangle + \sqrt{\frac{1}{j}}|j\rangle$ converges to the ground state as $j \to \infty$, while the expectation of N converges to 1. This means that [B]COBS is not guaranteed over QSTO:H.

The fundamental reason is that convergence of the probability distribution alone does not guarantee the convergence of the mean for an unbounded random variable. While this is also true in classical probability, one can, in principle, put a cutoff on all variables and work only with joint probability distributions in that compact region. Quantum mechanics does not allow that, so the topology (i.e. what constitutes small changes of preparations) has to be consistent with the choice of defining observable for each system.

Let us now show that [B]WDQ is not guaranteed over QSTO:H. Consider a quantum system described by a single spatial degree of freedom. Let x be the position variable used by the first observer, $\mathcal{H}_x = L^2(\mathbb{R}, dx)$ be the corresponding Hilbert space and X be the corresponding position operator. As shown in 3, $f(x) = \tan\left(\frac{\pi}{2}\operatorname{erf}(x)\right)$ is a smooth bijection over $\mathbb{R}$, and therefore we can let $y = f(x)$ be the position variable for a second observer, $\mathcal{H}_y = L^2(\mathbb{R}, dy)$ be the corresponding Hilbert space, Y be the corresponding position operator and U be the unitary transformation that represents the coordinate transformation. That is, $|\phi(y)\rangle = U|\psi(x)\rangle$.

Now, recall that a normalized state $|\psi\rangle$ is in the domain of a self-adjoint operator O if and only if $\|O\psi\|^2 = \langle O\psi|O\psi\rangle = E[O^2|\psi] < \infty$. That is, if and only if the second moment, and therefore the variance, is finite. As shown in 3, if we pick the Gaussian wave packet $\psi(x) = \sqrt{\frac{e^{-x^2}}{\sqrt{\pi}}}$ in the coordinates of the first observer, it is in the domain of X and therefore has finite variance. However, this transforms to $\phi(y) = \sqrt{\frac{1}{\pi(y^2+1)}}$ for the second observer, which is not in the domain of Y, and therefore its second moment diverges. The same state, then, has finite variance in position for the first observer while it has undefined variance in position for the second observer. Therefore [B]WDQ is not guaranteed over QSTO:H.

The problem is that unbounded operators have only a dense domain in a Hilbert space, and it is therefore possible to have two position variables with different domains. While coordinate transformations may produce similar problems in classical mechanics, one can again work with joint probability distributions with compact support, which bound all moments.

Lastly, we show that [B]STDIST is not guaranteed over QSTO:H. Consider two systems with different numbers of continuous degrees of freedom n and m. Under QSTO:H, these would be represented by $L^2(\mathbb{R}^n)$ and $L^2(\mathbb{R}^m)$ respectively. By 4, these are isomorphic, which means that there exists a unitary $U : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^m)$ between the two representations. This induces bijections between rays, density operators, and self-adjoint operators. The Born rule and the entropy are preserved. This means that two systems with different numbers of degrees of freedom can share the same mathematical representation. Therefore [B]STDIST is not guaranteed over QSTO:H.

This third failure is not related to the problems of continuity, as Schwartz spaces present the same problem. □

5 Additional remarks

There are other problems of a more practical nature. We highlight them for completeness.

Point-wise values not well-defined. The wave function $\psi(x)$ is not defined at each point. A vector of L^2, in fact, is the equivalence class of all such wave-functions that behave the same in the inner product. If $\psi(x)$ and $\phi(x)$ are different over a measure zero set, then the two wave functions are equivalent. This means that the values at each point of $\psi(x)$ are not uniquely defined. This goes against the physical intuition of interpreting $|\psi(x)|^2$ as the probability density, as that is conceptually uniquely defined.

Commutator relationships must be restricted. Commutation relationships like $\frac{[X,P]}{\imath\hbar} = I$ are not technically correct because of operator domains. Operators on the left, in fact, are not defined on the whole of $\mathcal{H}$ as they are unbounded. However, the identity I is well-defined over the whole space. Therefore, the relationship is not literally true in a Hilbert space, and must be restricted to the appropriate domain.

Schrödinger equation not well-defined. Note that the Schrödinger equation is not necessarily defined on all functions. In fact, functions in the Hilbert space are not necessarily differentiable and therefore the momentum operator may not be well-defined. Similarly, $V(x)\psi(x)$ may not be a square-integrable function. Therefore, the equation is valid only on the domain of H. If H is unbounded, this domain differs from the whole space.

6 Conclusion

It should be understood that all the issues presented here are failures of the mathematical modeling, not failures of physicists to correctly apply the mathematical model. Therefore, the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements.